23AID302 — BIG DATA ANALYTICS | 0TH REVIEW

MediSpark

A Scalable Multi-Disease Clinical Risk Prediction Engine

Apache Spark · Hadoop HDFS · Scala · Spark MLlib · Docker

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23AID302 — Big Data Analytics

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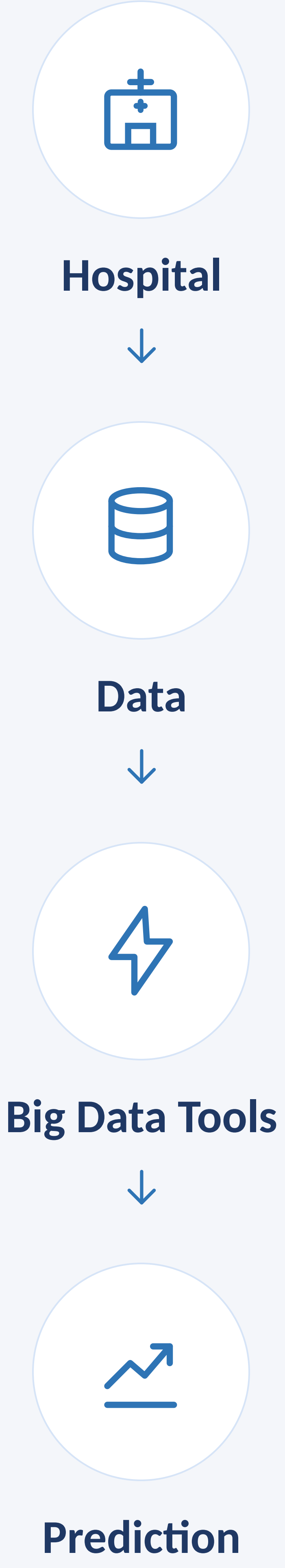
INTRODUCTION

Healthcare Meets Big Data

- The healthcare sector generates millions of patient records daily from hospitals, wearables, labs, and EHRs
- Traditional systems cannot process this volume, velocity, and variety of data in real time
- Chronic diseases — heart disease, diabetes, stroke, lung cancer — cause 70%+ of global deaths annually
- Early detection through data-driven prediction can save millions of lives
- Big Data technologies like Apache Spark and Hadoop HDFS now enable scalable, real-time healthcare analytics



DOMAIN IMPORTANCE

Why This Domain Matters

Scale

500 million TB of health data generated daily worldwide

Speed

ICU monitors generate real-time vitals every second

Complexity

Multi-format data — labs, images, notes, wearables

Value

Predictive analytics can cut misdiagnosis by 30%

Source: WHO Global Health Observatory & IDC Data Age 2025 Report

Problem Statement

Existing healthcare prediction systems are designed for a single disease, operate on static datasets, and cannot process the massive volume, velocity, and variety of modern patient data in real time — leaving hospitals without a scalable, unified, multi-disease early-warning system.

- ✗ Single-disease models — no unified multi-disease prediction exists
- ✗ Batch-only systems — cannot score new patient records in real time
- ✗ No veracity handling — messy, missing, inconsistent clinical data ignored

Literature Review — Related Work

Paper	Year	What They Did	Key Gap
Ed-daoudy et al. (Springer)	2023	Real-time heart disease prediction using Twitter + Kafka + Spark MLlib	Single disease only. No HDFS. No data quality pipeline.
Ebada et al.	2022	Stroke prediction using Spark MLlib batch processing	No streaming. No data variety. Single disease.
Kiran et al.	2020	Heart disease streaming prediction with Spark Streaming + MLlib	No HDFS. No ML Pipeline API. No veracity handling.
Ed-daoudy & Maalmi (Springer)	2019	IoT + Kafka + Cassandra + Spark for multi-stream disease monitoring	No HDFS. No Scala. Overly complex. No veracity.

All 4 papers address only 1–2 aspects. None provides a complete, unified, reproducible Big Data healthcare pipeline.

Research Gap

What Exists

- ✗ Single-disease prediction models only
- ✗ No distributed HDFS-based storage
- ✗ No dedicated data veracity/quality pipeline
- ✗ Streaming OR batch — never both together
- ✗ Complex infrastructure (Kafka, Cassandra) not reproducible
- ✗ No Scala-based unified ML Pipeline API implementation

What MediSpark Provides

- ✓ 4 diseases predicted simultaneously in one pipeline
- ✓ Full HDFS storage with 3× block replication
- ✓ Dedicated veracity pipeline — IQR, null handling, dedup
- ✓ Both batch training AND real-time Structured Streaming
- ✓ Simple Docker-based setup — fully reproducible
- ✓ Scala Spark ML Pipeline API — production-grade architecture

ARCHITECTURE

Proposed System — MediSpark Pipeline

Datasets (Kaggle/UCI) → HDFS Storage (3 DataNodes) → Spark Data Cleaning (Scala) → Feature Engineering → Spark MLlib Training (RF + LR) → Structured Streaming → Risk Score Output

- Processes 4 diseases — Heart, Diabetes, Stroke, Lung Cancer — in parallel
- Scores new patients in real time every 5 seconds via Spark Streaming
- Outputs: Patient ID | Disease | Risk Level (High/Medium/Low) | Confidence %

COMPARISON

How MediSpark Improves Over Existing Work

Existing Systems	MediSpark
1 disease predicted	4 diseases simultaneously
Python on single machine	Scala on distributed Spark cluster
Local CSV storage	HDFS with 3× replication
No streaming	Spark Structured Streaming — live scoring
Drop nulls only	Full veracity pipeline — IQR, dedup, validation
Individual models	Spark ML Pipeline API — modular, reusable
Complex infra (Kafka, Cassandra)	Simple Docker Compose — reproducible

CONTRIBUTION

What is New in MediSpark

Multi-Disease Parallelism

One unified Spark pipeline predicts 4 chronic diseases simultaneously. No existing student project or paper does this in a single framework.

Real-Time Streaming Layer

Spark Structured Streaming scores new patient records every 5 seconds. Batch + Streaming in the same pipeline.

Veracity-First Design

Dedicated data quality stage: IQR outlier removal, null imputation, duplicate detection, schema validation + quality report to HDFS.

Production-Grade Scala ML Pipeline

Built with Spark ML Pipeline API in Scala — the same architecture used at Netflix, Uber, LinkedIn — not just a notebook experiment.

OBJECTIVES

Objectives of the Project

1. To design a distributed data storage and ingestion pipeline using Hadoop HDFS and Docker.
2. To implement a data quality and veracity pipeline in Scala for cleaning and validating multi-format healthcare data.
3. To engineer feature transformation pipelines using Spark DataFrames and the ML Pipeline API.
4. To train and evaluate multi-disease risk prediction models using Spark MLlib (Random Forest + Logistic Regression).
5. To implement real-time patient risk scoring using Spark Structured Streaming.
6. To evaluate model performance using AUC-ROC, Precision, Recall, and F1-Score metrics stored in HDFS.

Tools, Technologies & Datasets

Tools & Technologies

Tool	Purpose
Apache Spark 3.5	Distributed data processing & ML
Hadoop HDFS	Distributed fault-tolerant storage
Scala	Primary pipeline implementation language
Spark MLlib	Model training — RF & Logistic Regression
Spark ML Pipeline API	Modular feature + model pipelines
Spark Structured Streaming	Real-time patient risk scoring
Docker & Docker Compose	Reproducible multi-node cluster setup
Jupyter / Zeppelin	Exploratory data analysis & visualization

Datasets

Dataset	Size	Source
Heart Disease UCI	303 records	UCI ML Repository
Pima Indians Diabetes	768 records	Kaggle
Stroke Prediction Dataset	5,110 records	Kaggle
Lung Cancer Dataset	1,000 records	Kaggle
Synthetic Streaming Feed	Generated	Custom Scala generator